

Tab 2

Inspired by the key role Systems Engineering plays in developing complex systems and by the emerging directions of Industry and Society 5.0, this conference addresses how we develop complex systems that demand seamless integration of technical, organizational, and human elements. This conference advances the successful practice of human-centered systems engineering.

Tab 1

43 - CASA: A Simple Tool to Ease the Mind of Systems Engineers

Modern engineering projects are increasingly complex, involving multiple stakeholders, interdependent systems, and high uncertainty. Teams often struggle to maintain clarity, rhythm, and balance. Although established systems engineering methods exist, applying them in ways that fit real-world project dynamics remains difficult.

This tutorial introduces CASA, a mindset tool designed to help practitioners navigate complexity, sustain balance, and maximize stakeholder value. CASA is represented as a physical token of three stacked discs. The bottom disc represents organizational project-enabling processes. The top disc represents technical processes, from business and mission analysis through design, verification, and disposal. The middle disc represents technical management and agreement processes. Atop the three discs sits a cone representing quality characteristics — affordability, interoperability, manufacturability, adaptability, and more. The discs and cone rotate independently, embodying the iterative nature of complex projects and supporting balance throughout.

You will engage with practical exercises to experience CASA in action: applying a structured mindset to complex projects, aligning organizational and technical processes, integrating quality characteristics explicitly, and using iterative cycles to refine solutions. CASA is a physical summary of the INCOSE Systems Engineering Handbook, and a practical entry point for both new and experienced practitioners.

Select

17 remaining

29/08/26

74 - Artificial Intelligence & SysML v2 as the Backbone of Generative MBSE Architectures

Creating initial system architectures in MBSE remains time-consuming and expertise-driven, slowing early design exploration. Recent advances in AI, particularly large language models, offer the potential to accelerate these early stages significantly. This tutorial presents a practical, end-to-end approach for combining AI with SysML v2 to rapidly bootstrap and evolve system architectures. In the first part, you will learn how to input a high-level system description into AI platforms and generate an initial SysML v2 architecture in textual notation, covering requirements, structure, and behavior. The generated model can be imported directly into a SysML v2-compliant MBSE tool as a starting point for further development.

In the second part, the tutorial demonstrates how built-in AI capabilities within MBSE tools and native modeling features can expand, refine, and validate the architecture. Critical considerations – model correctness, traceability, and the engineer's role in guiding AI-generated content – are addressed throughout. By the end of the session, you will understand how to integrate AI into SysML v2-based MBSE workflows, improving efficiency while maintaining engineering rigor.

Select

35 remaining

30/08/26

Synopsys Tutorial - Engineering Better Decisions: A Hands-on Journey Through Connected Digital Engineering

More information to be provided soon. What does it take to develop a modern electro-mechanical system while maintaining traceability, managing complexity, and making confident engineering decisions?

In this interactive workshop, participants will explore the development of an ePowertrain system using a connected digital engineering approach powered by Model-Based Systems Engineering. Working through realistic engineering scenarios, attendees will experience how requirements, system architecture, simulation, software development, verification, and safety analysis can be integrated within a continuous digital thread. See how engineering data flows seamlessly across disciplines, enabling teams to collaborate more effectively, identify issues earlier, evaluate trade-offs with greater confidence, and accelerate development while preserving full lifecycle traceability.

Whether you are a systems engineer, software engineer, architect, or engineering leader, this workshop will provide practical insight into how connected digital engineering transforms the way complex products are developed and verified.

Tab 3

These sources explore the evolution of human-centric manufacturing and human-machine teaming (HMT), specifically focusing on how complex industrial systems can better integrate people and technology. The first source proposes a systems engineering framework designed for the transition to Industry 5.0, emphasizing visualization and the use of "personas" to manage risk and complexity in new battery production facilities. This research advocates for moving beyond equipment-focused design to prioritize the practical and emotional needs of human workers. Complementing this, the second source provides a broad interdisciplinary survey of HMT across sectors like defense and healthcare, examining how trust, ethics, and shared mental models drive effective collaboration. It outlines a detailed taxonomy and architecture for HMT, identifying critical success factors such as team resilience, accountability, and the calibration of human-AI trust. Together, these texts argue that the future of advanced manufacturing and autonomous systems relies on synergistic partnerships where machine precision supports human creativity and oversight.

Today • 7:48 AM

How can systems engineering reduce complexity and risk in manufacturing?

Thoughts

Analyzing the Information

I'm carefully reviewing the provided information to determine how systems engineering can reduce manufacturing complexity and risk, drawing key findings.

Reviewing the Framework

I am now focusing on how Model-Based Systems Engineering (MBSE) and visual system models reduce complexity and risk for new products.

Considering the Standards

I'm now investigating relevant standards to identify structured approaches for managing project lifecycles and addressing uncertainties.

Exploring Predictive Modeling

I'm now focusing on how ****digital twins**** and predictive analysis provide real-time feedback, enabling faster assessments of design choices before production [156, 243, 248, 252, 337]. This creates a shared system view to overcome ****knowledge silos**** [11, 12, 17, 26]. I believe we can now explore the use cases in depth.

Deepening the Analysis

I'm now revisiting citations to ensure each point is thoroughly supported, especially regarding digital twins and complexity definitions.

Focusing on Integration

I'm now examining Human-Systems Integration (HSI) to reduce errors and improve collaboration across teams, for a clearer plan.

Systems engineering reduces complexity and risk in manufacturing by shifting design from fragmented, document-based methods to structured, visual, and model-centric frameworks across the system life cycle.

1. Early Visual Modeling and Cross-Functional Alignment

Complexity in manufacturing stems not only from technical equipment, but from interactions between people, functional departments, and changing organizational environments. Traditional manufacturing projects often suffer from knowledge silos, ambiguous responsibilities, and poorly understood specification documents.

Systems engineering addresses this by introducing visual system models and Concept of Operations (ConOps) during the early concept and specification phases:

- Structuring early discussions: Visualizing the whole system overview allows cross-functional teams to look at the same picture, ask the right questions early, and make system complexities tangible.
- Shared ownership: Bringing representatives from across the production lifecycle together fosters collective ownership of the entire system rather than isolated hand-offs.

2. Model-Based Systems Engineering (MBSE) and Digital Threads

Integrating Model-Based Systems Engineering (MBSE) with Digital Twins, Cyber-Physical Systems (CPS), and Manufacturing Operations Management (MOM) establishes a continuous digital thread:

- Synchronizing resources (4Ms): Frameworks like the Dynamic Digital Thread Framework (DDTF) use MBSE to align Man, Machine, Material, and Method, reducing conversion costs and eliminating non-value-added activities.
- Evaluating design alternatives: High-fidelity modeling and multi-disciplinary simulation allow engineers to assess an order of magnitude more alternative system architectures quickly and thoroughly before physical implementation.

3. Predictive Risk Management & Digital Twins

Systems engineering structures uncertainty management throughout the system life cycle (such as via ISO/IEC/IEEE 15288 standards):

- Pre-deployment risk identification: By leveraging digital twins and enhanced quality control frameworks (e.g., digital DMAIC), teams can simulate operational flows, run anomaly detection algorithms, and identify failure modes during design rather than during plant bring-up.
- Asset reliability: Integrating predictive analytics continuously monitors machine health, boosting Mean Time Between Failures (MTBF) and minimizing unplanned downtime.

4. Human-Systems Integration (HSI) and Ergonomic Risk Reduction

A central pillar of modern human-centric systems engineering (Industry 5.0) is treating the human operator as an explicit part of the system:

- Mitigating operational errors: Incorporating personas, ergonomic modeling, and Human-Systems Integration (HSI) early reduces cognitive overload, attention allocation issues, and mode errors on the factory floor.
- Shared Mental Models (SMM): Designing transparent interactions between workers and automated or AI systems creates shared mental models, improving situational awareness and mitigating risks associated with sudden automation failures.

💡 **Next Step Idea:** Would you like to explore how to create a structured framework or proposal for implementing an MBSE and Digital Twin workflow in your specific manufacturing process?

How can MBSE and Digital Twins be structured for my process?

Thoughts

To structure Model-Based Systems Engineering (MBSE) and Digital Twins for a manufacturing process, your framework must bridge high-level system architecture with real-time factory execution.

A structured approach integrates MBSE, Digital Twins, Cyber-Physical Systems (CPS), and Manufacturing Operations Management (MOM) across four core stages:

1. Early Architecture & Concept Definition (MBSE & ConOps)

Before deploying digital twins or hardware on the shop floor, use MBSE to establish early system models and functional specifications:

- **Technical Process & ConOps Assessment:** Begin by assessing technical processes and formulating a Concept of Operations (ConOps) in a collaborative setting.
- **Cross-Functional Visual Modeling:** Build SysML or visual system models to replace document-based engineering. Visualizing the system overview allows engineering, operations, and management teams to evaluate design trade-offs and alternative architectures early in the lifecycle.
- **Human-Systems Integration (HSI):** Embed operator personas, ergonomics, and practical/emotional operator needs directly into early MBSE models to ensure human factors are designed into the workflow from the start.

2. Multi-Layer Information Architecture (The Dynamic Digital Thread)

To ensure continuous information flow, connect MBSE to plant operations via a multi-layered architecture like the Dynamic Digital Thread Framework (DDTF):

1. **MBSE Layer (Engineering Intent):** Defines requirements, functional architectures, and system behavior models.
 2. **Digital Twin Layer (Analytical Intelligence):** Maintains a continuous dynamic virtual mirror of physical assets, running physics-based and data-driven simulations.
 3. **Cyber-Physical Systems (CPS) & IoT Layer (Physical Sensing & Actuation):** Provides IoT connectivity for real-time data ingestion, status monitoring, and adaptive control signals between physical equipment and digital models.
 4. **MOM / MES Layer (Executional Control):** Manages floor-level scheduling and execution, synchronizing the 4Ms (Man, Machine, Material, and Method) to lower conversion costs and eliminate non-value-added activities.
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3. Human-in-the-Loop & Teaming Architecture

Structure how human operators interact with automated systems by incorporating explicit human-machine teaming components:

- Shared Mental Models (SMM): Model both the task requirements (goals, processes, workload) and the team roles (capabilities of both human operators and machine/AI agents) inside your system architecture.
- Adaptive Task Allocation & Feedback Loops: Establish real-time feedback loops that monitor human cognitive workload, situational awareness, and system trust, allowing task assignments to shift dynamically during operational disruptions.

4. Operational Improvement & Phased Rollout

Integrate MBSE and Digital Twins into continuous process optimization using an Enhanced Digital DMAIC approach alongside a phased rollout:

- Digital DMAIC Cycle:
 1. *Define & Measure*: Map stakeholders and instrument physical equipment with IoT sensors feeding live baseline data into the Digital Twin.
 2. *Analyze & Improve*: Use AI anomaly detection and process mining on the twin to diagnose root causes and simulate changes (via AR/VR or dynamic models) before physical implementation.
 3. *Control*: Deploy adaptive controls and interactive operator dashboards for continuous monitoring.
- Phased Implementation Roadmap:
 1. Vision & Alignment: Align executive leadership on human-centric goals and scope.
 2. Readiness Assessment: Evaluate technical infrastructure, data architecture, and workforce skill gaps.
 3. Strategy & Roadmap Design: Define priorities, target KPIs (OEE, FPY, MTBF), and governance structures.
 4. Pilot & Capability Building: Test cobots, digital twin modules, or MBSE workflows in a single production cell while upskilling staff.
 5. Integration & Scaling: Scale validated digital thread models across additional lines and facilities.
 6. Continuous Innovation: Utilize real-time analytics to sustain optimizations and maintain systemic resilience.



Next Step Idea: Would you like to create a tailored Implementation Roadmap or Report summarizing these phases and KPIs for your project team?

How can personas help include human factors?

Thoughts

In systems engineering and Industry 5.0 manufacturing, personas play a crucial role in shifting design away from purely equipment-centric methods toward human-centric socio-technical systems.

Personas help incorporate human factors in several concrete ways:

1. Capturing Practical and Emotional Needs

Traditional manufacturing engineering focuses primarily on technical equipment and physical hardware. Incorporating personas allows cross-functional design teams to systematically model and understand both the practical needs (such as task workflows and physical ergonomic constraints) and the emotional needs (such as job satisfaction, agency, and cognitive comfort) of operators working within the system.

2. Driving Early Human-Centric Requirements

In conventional document-based design, human factors are often treated as an afterthought or added during final implementation. By using personas during the early Concept of Operations (ConOps) and Model-Based Systems Engineering (MBSE) specification phases, engineers can formulate high-level human-centric requirement specifications alongside technical equipment requirements. This ensures ergonomics, safety, and cognitive workload boundaries are designed into the system architecture from the outset.

3. Fostering Shared System Ownership across Teams

Engineering projects frequently suffer from knowledge silos where departments focus strictly on delivering their specific hardware component. Using personas in visual modeling workshops gives project members a shared, empathetic understanding of the operator's end-to-end experience. Validation studies show that project teams find it rewarding to view the production facility as a co-owned sociotechnical system rather than just a collection of disconnected machines.

4. Structuring Human-Machine Teaming and Task Allocation

In Industry 5.0 environments featuring AI, digital twins, and collaborative robots (cobots), personas help define clear roles, capabilities, and decision boundaries for human operators. This prevents cognitive overload, mitigates mode errors, and ensures technology acts as an empowering cognitive assistant that preserves human agency rather than a restrictive automated monitor.

 **Next Step Idea:** Would you like to draft a sample operator persona or persona template tailored for your factory floor to help guide your early MBSE requirements modeling?

Draft an operator persona for my factory floor

Thoughts

Here is a sample operator persona designed for early Model-Based Systems Engineering (MBSE) and Concept of Operations (ConOps) requirements modeling. Using personas during early concept phases shifts project focus from purely equipment-centric design to an integrated, human-centric socio-technical system. It

systematically captures both the practical needs (physical ergonomics, task workflows) and emotional needs (cognitive agency, trust, job satisfaction) of factory floor operators.

Operator Persona: "Elena Vance" — Assembly & Cobot Teammate

- Role: Senior Cell Assembly Operator / Hybrid Station Technician
- Experience: 8 years in powertrain manufacturing; 1 year working alongside collaborative robots (cobots) and AI decision-support tools
- Workplace Context: High-variability, semi-automated assembly line handling novel battery module integration

1. Practical Needs (Operational & Physical Ergonomics)

- Intuitive HMI Displays: Requires clear, visual, real-time status updates at the workstation interface without navigating multi-layered text menus.
- Physical Ergonomics & Assistive Automation: Needs cobot automation to take over heavy lifting, awkward reaching, and repetitive precision tasks to reduce physical fatigue.
- Clear Task Workflows & Instructions: Needs dynamic guidance when switching product variants to minimize assembly errors.

2. Emotional & Psychological Needs (Agency, Trust, & Culture)

- Preserved Human Agency: Wants to remain the ultimate decision-maker on the line, acting as a system governor rather than feeling controlled or monitored by automated tools.
- Transparent System Behavior (Trust Calibration): Needs to understand *why* an AI decision tool or digital twin recommends a process change or flagged a defect before taking corrective action.
- Shared Ownership & Recognition: Values being involved in early system design discussions so her practical floor experience shapes how equipment is configured.

3. Pain Points & Operational Frustrations

- Black-Box Automation & Mode Errors: Experiences frustration when cobots or automated tools stop unexpectedly or switch operating modes without clear diagnostic feedback on screen.
- Cognitive Overload During Line Disruptions: Struggles when multiple sensor alarms fire simultaneously during a bottleneck, causing attention allocation stress.
- Rigid Task Reallocation: Frustrated by fixed automation schedules that fail to account for worker fatigue or unexpected physical strain during high-throughput shifts.

4. Shared Mental Model (SMM) Expectations

- Task Model Alignment: Expects the system to maintain a mutual understanding of current production goals, assembly step sequences, and quality standards.
- Team Model Alignment: Needs explicit role clarity between what she handles (judgment, flexible handling, quality verification) versus what machine agents handle (speed, repetitive positioning, sensor logging).

Translating Persona Needs into Early MBSE System Requirements

To feed Elena's profile into your SysML architecture and early requirements specification:

1. R-HSI-01 (Visual Transparency): *The Human-Machine Interface (HMI) shall display real-time machine intent and fault diagnostics in graphical visual format within 1 second of an event, maintaining operator situational awareness.*
2. R-HSI-02 (Adaptive Task Allocation): *The production system shall dynamically adjust task sharing between operator and cobot based on real-time cognitive workload and physical cycle constraints.*
3. R-HSI-03 (Human Agency Override): *The control architecture shall provide the human operator with unambiguous manual override authority over AI process recommendations at all operational nodes.*

 Next Step Idea: Would you like to map this persona to a specific station on your factory floor (e.g., quality inspection, sub-assembly, or material handling) to refine the exact SysML requirement parameters?

yes

Thoughts

Mapping Elena Vance's operational role to specific factory floor stations translates human factors, cognitive workload, and trust requirements into formal SysML requirement parameters.

Below is a detailed mapping of Elena's persona across three key manufacturing stations:

1. Station A: Semi-Automated Battery Module Sub-Assembly

- Workflow & Teaming Context: Elena works alongside a collaborative robot (cobot) handling heavy battery cell insertion, thermal interface material (TIM) dispensing, and high-precision mechanical fastening.
- Human Factors Focus: Reducing physical fatigue from repetitive lifting, preventing mode errors during variant switchovers, and enabling dynamic task reallocation.

SysML Requirement Parameters

- `sysml:Requirement [R-HSI-SUB-01]` — Ergonomic Assistive Payload Control

- *Text:* The sub-assembly station cobot subsystem shall automatically execute all lifting tasks exceeding 5.0 kg and perform high-precision positioning to maintain operator musculoskeletal strain within allowable ergonomic limits.
- *Verification Method:* Test & Biomechanical Simulation (EMG / Ergonomic Modeling).
- *Traced Persona Need:* Physical Ergonomics & Assistive Automation.
- **sysml:Requirement [R-HSI-SUB-02]** — Adaptive Task Reallocation
 - *Text:* The station control architecture shall dynamically adjust assembly task sharing between Elena and the machine agent based on real-time cycle latency and physical fatigue metrics without stopping line throughput.
 - *Verification Method:* Demonstration & Human-in-the-Loop Simulation.
 - *Traced Persona Need:* Mitigation of Rigid Task Reallocation & Physical Strain.

2. Station B: AI-Assisted Quality Inspection & Defect Verification

- Workflow & Teaming Context: Machine vision and deep learning algorithms perform initial surface defect scanning on assembled battery modules, while Elena conducts final expert validation and root-cause verification.
- Human Factors Focus: Calibrating trust in AI predictions, preventing cognitive overload from false-positive alarms, and maintaining human agency as the final quality authority.

SysML Requirement Parameters

- **sysml:Requirement [R-HSI-INSP-01]** — XAI Decision Transparency & Bounded Latency
 - *Text:* The AI vision inspection system shall display defect classification confidence scores and visual heatmap bounding boxes on Elena's HMI within 0.5 seconds of scan completion to support explainable decision-making.
 - *Verification Method:* Inspection & HMI Response Time Measurement.
 - *Traced Persona Need:* Transparent System Behavior & Trust Calibration.
- **sysml:Requirement [R-HSI-INSP-02]** — Human Agency Override & Feedback Logging
 - *Text:* The inspection system shall provide an unambiguous, single-touch HMI override allowing Elena to overturn AI defect classifications, logging operator rationale to refine the digital twin's model.
 - *Verification Method:* Test & Software Audit.
 - *Traced Persona Need:* Preserved Human Agency & System Governance.

3. Station C: Material Handling & AMR Line Replenishment

- Workflow & Teaming Context: Autonomous Mobile Robots (AMRs) deliver raw cell kits and assembly fasteners to Elena's workstation.
- Human Factors Focus: Maintaining spatial situational awareness, avoiding workspace congestion, and preventing mode errors during manual material takeover.

SysML Requirement Parameters

- **sysml:Requirement [R-HSI-LOG-01]** — Spatial Awareness & Intent Visualization
 - *Text:* The AMR replenishment system shall communicate vehicle approach path, arrival ETA, and docking orientation via workstation floor projections and HMI status graphics at least 10 seconds prior to cell arrival.
 - *Verification Method:* Demonstration & Field Test.
 - *Traced Persona Need:* Spatial Situational Awareness & Collision Prevention.
- **sysml:Requirement [R-HSI-LOG-02]** — Mode Error & Takeover Prevention
 - *Text:* The material transfer HMI shall explicitly indicate system operational mode (Autonomous Delivery vs. Manual Loading Lock) with distinct color coding and audible confirmation to eliminate mode confusion.
 - *Verification Method:* Test & Task Load Analysis (NASA-TLX).
 - *Traced Persona Need:* Elimination of Mode Errors & Black-Box Automation.

Connecting to the Broader MBSE Architecture

In your digital modeling tool (such as SysML v2, IBM Rhapsody, or Enterprise Architect):

1. Requirements Diagrams: Link these **sysml:Requirement** blocks to persona attributes (**Elena_Vance:Persona**) via **«trace»** or **«satisfy»** relationships.
2. Activity & Sequence Diagrams: Embed human decision nodes and HMI feedback loops directly into operational process flows alongside machine actions.
3. Parametric Blocks: Connect human performance constraints (e.g., maximum cognitive load, HMI response delays) with station cycle time equations to simulate human-system performance before physical plant bring-up.

 **Next Step Idea:** Would you like to generate a complete requirements traceability matrix or document report compiling these SysML blocks for your engineering team?

Generate a Requirements Traceability Matrix

Thoughts

I have generated and published a complete Requirements Traceability Matrix (RTM) for your human-centric manufacturing system.
The deliverable is available in your Studio panel in two formats:

- 1. Excel Workbook (requirements_traceability_matrix.xlsx): A fully formatted spreadsheet with an Executive Summary & TOC tab, automated formula integrity checks, color-coded input/calculation cells, and auto-filters for engineering teams.
- 2. PDF Document Report (requirements_traceability_matrix.pdf): A publication-quality landscape document designed for executive reviews and cross-functional design gates.

Matrix Overview & Traceability Mapping

The matrix bridges Elena Vance's operator persona needs with formal SysML requirement specifications, the 4-layer Dynamic Digital Thread Framework (DDTF), and quantitative verification KPIs across 12 core requirements:

Req ID	Stakeholder / Persona Need	Functional Domain	Digital Thread Layer	Target KPI / Acceptance Criteria	Grounding Citation
REQ-H SI-001	Physical Ergonomics & Assistive Automation	Station A: Sub-Assembly	CPS / IIoT Actuation Layer	0 manual lifts >5 kg; EMG strain <20% MVC	Hagström & Bergsjö (2024); Gomaa (2026)
REQ-H SI-002	Adaptive Task Reallocation & Fatigue Balance	Station A: Sub-Assembly	Digital Twin & SMM Engine	NASA-TLX workload <40/100; Line Rate ≥98%	Chen et al. (2018); Dubois & Le Ny (2020)

REQ-H SI-003	Transparent Vision XAI & Bounded Latency	Station B: Quality Inspection	Digital Twin / XAI Layer	HMI Latency <0.5s; XAI confidence displayed	Chen et al. (2018); Chen (2017)
REQ-H SI-004	Preserved Human Agency & Override Authority	Station B: Quality Inspection	MOM / MES & Agency Core	100% manual override success rate; Audit log created	Farivar et al. (2025); Gomaa (2026)
REQ-H SI-005	Spatial Situational Awareness & Intent Projection	Station C: AMR Logistics	CPS / IIoT & AMR Fleet Controller	Vehicle arrival ETA notice ≥ 10 s; 0 collisions	Chen et al. (2018); Schelble et al. (2022)
REQ-H SI-006	Mode Error & Takeover Prevention	Station C: AMR Logistics	HMI / CPS Interaction Layer	Mode error rate = 0; State confusion = 0	Chen et al. (2018); Parasuraman et al. (2008)
REQ-H SI-007	Shared Mental Model (SMM) Alignment	System-Wide Governance	MBSE Model & SMM Engine	Pathfinder C similarity score >0.85; 0 goal conflicts	Chen et al. (2018); Andrews et al. (2023)

REQ-H SI-008	Dynamic Digital Thread 4Ms Alignment	System-Wid e Governance	MOM / MES & Digital Thread	Conversion cost reduction ≥12%; Non-value waste <5%	Silva Sprock (2026); Gomaa (2026)
REQ-H SI-009	Predictive Maintenance & Anomaly Detection	System-Wid e Governance	Digital Twin & Analytics	MTBF increase ≥25%; Unplanned downtime -30%	Silva Sprock (2026); Gomaa (2026)
REQ-H SI-010	Early Concept Visual ConOps Modeling	System-Wid e Governance	MBSE Architecture & ConOps	100% early requirement validation; 0 spec defects	Hagström & Bergsjö (2024); Boy & Kennedy (2021)
REQ-H SI-011	Ethical AI & Data Governance	System-Wid e Governance	Ethical Governance Layer	100% compliance with ISO 27001 & EU AI Guidelines	Gomaa (2026); Farivar et al. (2025)
REQ-H SI-012	Continuous Operator Reskilling & Training	System-Wid e Governance	HMI / AR/VR Training Subsystem	Variant onboarding time -40%; First Pass Yield ≥99%	Gomaa (2026); Farivar et al. (2025)

💡 Next Step Idea: Would you like to adjust any specific quantitative parameters (such as HMI response latency thresholds or NASA-TLX target scores) or add custom station parameters for your facility?